10.1 Newton's Cradle

WORKSHEET

The question

STEP II 1990, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q14

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself A marble of mass m moving at speed u hits a stationary marble of mass M, and kinetic energy is conserved. What are the velocities afterwards? What if $M = m$?

STAGE 1 Only the last two move

Ask yourself Suppose only A_n (velocity p) and A_{n+1} (velocity q) move afterwards. Write down the conservation laws. Which marble's motion constrains the sign of p?

STAGE 2 Three marbles

Ask yourself Now suppose only A_{n-1}, A_n and A_{n+1} move, with velocities a, b, c. What inequalities must a, b, c satisfy, given which marbles are at rest and which are behind which?

Ask yourself Square the momentum equation and compare it with the energy equation. What is left?

Pause and try Notice the structure of the argument. Squaring the momentum equation produces cross terms like $2ab$, and the energy equation has none. The difference measures how "spread out" the motion is. Can you see why the same argument rules out *any* outcome in which only $A_k, \ldots, A_{n+1}$ move with non-negative velocities, when $\lambda > 1$?

STAGE 3 Two marbles that can move

Ask yourself If the marbles moving forwards must all be at the far end, and that fails, perhaps one marble moves *backwards*. Which marble is free to move backwards?

What's really going on

Why one ball comes out

In a Newton's cradle of equal balls, suppose k balls leave together at speed w. Momentum gives $kw = u$ and energy gives $kw^2 = u^2$. Dividing, $w = u$ and $k = 1$. So among outcomes where the departing balls move together, only "one ball at the original speed" obeys both laws. Two balls at half speed carry the right momentum but only half the energy; one ball at twice the speed would have the right momentum only if it were half a ball.

But this does not settle the matter. With n balls there are n unknown velocities and only two conservation laws, so there are infinitely many outcomes that obey both (try finding one with three balls moving at different speeds). Which one happens depends on how the balls deform during contact. In the early 1980s Frank Herrmann and his colleagues showed that the familiar behaviour depends on the shock passing along the line *without spreading out*, which happens for hard steel balls with tiny gaps between them. With soft balls, or balls pressed tightly together, the cradle behaves less cleanly, and careful measurements show that the other balls always move a little.

Conservation laws and symmetry

Where do the conservation laws come from? In 1918 Emmy Noether proved a theorem that is one of the foundations of modern physics: every continuous symmetry of a physical system gives a conserved quantity. If the laws of physics are the same here as one metre to the left (symmetry under translation in space), momentum is conserved. If they are the same today as tomorrow (symmetry under translation in time), energy is conserved. If they are the same in every direction (rotational symmetry), angular momentum is conserved. You will meet Noether's theorem in a second-year course on Lagrangian mechanics; this chapter uses its consequences throughout.

Huygens, Wren and Wallis

In 1668 the Royal Society invited solutions to the problem of colliding bodies. John Wallis treated inelastic collisions, and Christopher Wren and Christiaan Huygens independently solved the elastic case. Huygens's key insight was to change frame: view the collision from a boat moving with the centre of mass, where two equal balls simply bounce back with their speeds reversed, and then transform back to the shore. He also noticed that a certain quantity, the sum of the masses times the squares of the speeds, was the same before and after, though it was another century before kinetic energy was recognised as a conserved quantity in its own right.

Try this next

1 **The same problem again.** Try STEP I 2006, Question 11 (step.maths.org/questions/06-s1-q11): a row of particles in a trough, with a heavier one at the end. Which final states are possible?

2 **Following the collisions.** Model the impact in part (iii) as a sequence of separate two-marble collisions, starting with A_0 hitting A_1. Follow the sequence through and show that it ends in the motion found in part (iii).

3 **Two balls in.** In a cradle of five equal balls, two balls are released together and hit the other three with speed u. Show that "two balls leave with speed u" is consistent with both conservation laws, and find another outcome that is.

4 **Light end.** If $\lambda < 1$, follow the sequence of two-marble collisions. Do any marbles other than A_n and A_{n+1} end up moving?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5. Conservation of momentum and energy, with many problems on collisions.
- **Frank Herrmann & Peter Schmälzle, "Simple explanation of a well-known collision experiment"**, *American Journal of Physics* 49 (1981). Why Newton's cradle works.
- **Dwight Neuenschwander, *Emmy Noether's Wonderful Theorem*** (Johns Hopkins University Press). An accessible account of the link between symmetry and conservation.

The full guided solution is in Section 10.1:
https://beyondstep.keithmcnulty.org/ch10/newtons-cradle.html